

Emmanuelle Coutu-Nadeau

emmanuelle@coutunadeau.com • www.emmanuellecoutunadeau.com

Education, Academic Awards & Scholarships

- Master of Science, Psychology – University of Montreal** (GPA: 4.2 / 4.3) **Sep 2024 – Present**
- Neuro-AI UNIQUE Excellence Scholarship (10,000 \$)
- Bachelor of Software Engineering – McGill University** (GPA: 3.7 / 4.0) **Sep 2018 – May 2022**
- MES Bourse de persévérance TI (3,000 \$)
 - Airbus Scholarship in Engineering (1,500 \$)
- Bachelor of Science, Psychology (Honours) – McGill University** (GPA: 3.8 / 4.0) **Sep 2015 – May 2018**
- First-Class Honours

Research Experience

- Master Student Researcher – Neurosciences of Early Development Lab (CHU Ste-Justine)** **Sep 2024 – Present**
- M.Sc. Thesis: Developed a ML approach to identify transdiagnostic behavioral profiles and associated EEG signatures in children and adolescents with neurodevelopmental conditions.
 - Conducted clinical EEG recordings with children presenting ASD and other complex neurodevelopmental conditions.
 - Database Standardization: Led a weekly initiative to modernize the lab's database architecture; evaluated database solutions and established unified standards for variable naming, data management, and code sharing.
 - Technical skills: Python (scikit-learn, MNE-Python, pandas), R, EEGLab, REDcap
- Research Assistant – Co-venture Lab (CHU Ste-Justine)** **Nov 2017 – Jul 2018**
- Assisted with data entry for a study on substance use and adolescent cognitive development.
- Honours Trainee – Martel Pain Lab (McGill University, MUHC)** **Oct 2016 – May 2019**
- B.Sc. Honours Thesis: Association between perceived injustice and prescription opioid misuse.
 - Conducted structured clinical interviews, data collection, and saliva sampling.

Professional Experience

- Product Owner – CareSimple** **Oct 2022 – Sep 2025**
- Coordinated clinical and technical teams for a liver transplant research project.
 - Translated client needs into technical specs for a digital health platform in the Swedish market.
 - Managed compliance documentation for Canadian and European health regulations.
 - Prototyped KPI reporting (Python, SQL) to support data-driven decisions.
- Software Engineer – CareSimple** **May 2020 – Oct 2022**
- Developed features for a web-based portal for clinicians to perform remote patient monitoring (React, TypeScript, Javascript).

Peer-reviewed Journal Articles

Coutu-Nadeau, E., Arutiunian, V., Martin, C.-O., Fontaine, V., Bélanger, A.-M., Jizi, K., Knoth, I. S., Hosein, A., Davoudi, S., Elsabbagh, M., Forgeot D'Arc, B., Jacquemont, S., & Lippé, S. (under review). Behavioral Clustering and Resting-State EEG Signatures of Transdiagnostic Neurodevelopmental Profiles. <https://doi.org/10.21203/rs.3.rs-10692457/v1>

Arutiunian, V., Davoudi, S., Knoth, S.I., **Coutu-Nadeau, E.**, Desjardins, J., O'Reilly, C., Hosein, A., Gomez, G.B., Martin, C.-O., Fontaine, V.K., Bélanger, A.M., Hébert-Lalonde, N., Rouleau, G., D'Arc, B.F., Elsabbagh, M., Jacquemont, S., Lippé, S. (under review). Auditory Sustained Brain Activity Links Genomic Variation to Language in Neurodevelopmental Disorders.

Verner M, Sirois A, **Coutu-Nadeau E**, Fournier C, Carriere J, Martel MO. The association between perceived injustice and opioid craving in patients with chronic pain. J Addict Med. 2022 Jul 21. <https://doi.org/10.1097/adm.0000000000001022>

Clinically Relevant Experience

Patient & Bedside Support Volunteer – CHU Sainte-Justine

Sep 2026 –

- Incoming, September 2026.

Crisis Responder – Kids Help Phone

Jul 2026 – Present

- Developed skills in text-based empathetic communication, collaborative safety planning, and real-time risk management.
- Completed intensive crisis intervention training focused on active listening, risk assessment, and suicide de-escalation frameworks for youth.

Pediatric Oncology, Cognitive Evaluator – CHU Ste-Justine

Sep 2025 – Present

- Administered standardized cognitive assessments to children and adolescents in oncology remission (WISC-V, WAIS-IV, WIPPSI-III, CMS, CVLT-C).
- Communicated with parents and practiced active listening to address concerns.

Mentor – Technovation Montreal

Jan 2024 – May 2026

- Mentored five groups of girls (ages 8–18) from January to May, guiding them through mobile app and business plan development to address a UN Sustainable Development Goal.
- Built trusted alliances with youth from diverse backgrounds, fostering self-confidence and encouraging engagement with technology as a tool for social good.
- Facilitated group dynamics and mediated interpersonal conflict within teams, while providing ongoing support to both youth and parents throughout the program.

Clinical Research Assistant (Substance use) – Montreal General Hospital (MUHC)

Oct 2016 – May 2019

- Conducted structured clinical interviews (SCID-5) with adult patients experiencing chronic pain and prescription opioid use disorders.
- Cultivated strong therapeutic alliances with a vulnerable adult clinical population navigating chronic pain and substance use.

Community Engagement

La Maison du Père – Board Member

Sep 2026 –

- Incoming, September 2026.

UNIQUE Center – Student Affairs Committee

Oct 2025 – Present

- Organized events and curated research highlights to support students at the intersection of Neuroscience and AI across Quebec.
- Participated in a panel discussion on the transition from industry to academia.

MAC Foundation – Event Ambassador

Jan 2025 – Apr 2026

- Promoted for “Les Printemps du MAC” 2025 and 2026 editions, an initiative supporting Montreal’s Museum of Contemporary Arts through fundraising events.

Pensionnat du Saint-Nom-de-Marie Foundation

Jan 2023 – Apr 2026

- Led and mentored a group of high school aged youth during a two-day bootcamp, facilitating structured leadership workshops and team-building activities.
- Spoke at Career Day events and helped organize annual panels with school alumni.
- Organized the 2013 cohort’s 10-year reunion.

Promoting Opportunities for Women in Engineering (POWE)

Feb 2019

- Led lab visits at The Neuro, introducing high-school girls to neuroscience and biomedical engineering.

Meals On Wheels “Santropol Roulant”

Nov 2016 – Jun 2022

- Supported community food security through meal preparation, home delivery for homebound individuals, and urban agriculture initiatives.

Published abstracts and conference presentations

Coutu-Nadeau E, Arutiunian V, Martin CO, Fontaine V, Bélanger AM, Jizi K, Knoth IS, Hosein A, Davoudi S, Elsabbagh M, Forgeot D'Arc B, Jacquemont S, Lippé S. Behavioral Clustering and Resting-state EEG signatures of Neurodevelopmental Profiles Using Machine Learning. Society for Psychophysiological Research; 2026 Oct (upcoming); Halifax, NS, Canada.

Coutu-Nadeau E, Arutiunian V, Martin CO, Fontaine V, Bélanger AM, Jizi K, Knoth IS, Hosein A, Davoudi S, Elsabbagh M, Forgeot D'Arc B, Jacquemont S, Lippé S. Behavioral Clustering and Resting-state EEG signatures of Neurodevelopmental Profiles Using Machine Learning. Child Neurology Society; 2026 Oct (upcoming); Montreal, QC, Canada.

Coutu-Nadeau E, Arutiunian V, Martin CO, Fontaine V, Bélanger AM, Jizi K, Knoth IS, Hosein A, Davoudi S, Elsabbagh M, Forgeot D'Arc B, Jacquemont S, Lippé S. Behavioral Clustering and Resting-state EEG signatures of Neurodevelopmental Profiles Using Machine Learning. 93e Congrès de l'Acfas (TACC Trainee Research Day on Autism and Neurodivergence) ; 2026 May 12; Trois-Rivières, QC, Canada.

Coutu-Nadeau E, Arutiunian V, Martin CO, Fontaine V, Bélanger AM, Jizi K, Knoth IS, Hosein A, Davoudi S, O'Reilly C, Rouleau G, Ernst C, Lucas-Tardif C, Scorah J, Couture M, Abadie P, Joobor R, Forgeot d'Arc B, Jacquemont S, Elsabbagh M, Lippé S. Resting-state EEG signatures of ASD/ADHD: toward identifying behavioral subgroups. Society for Psychophysiological Research Annual Meeting; 2025 Oct; Montreal, QC, Canada.

Kennedy G, Baran-Goldwax M, Pomerleau R, **Coutu-Nadeau E**, Knoth IS, Lippé S. Hormonal modulation of brain dynamics in menstruating adolescents and young adults: EEG insights across the menstrual cycle. Society for Psychophysiological Research Annual Meeting; 2025 Oct; Montreal, QC, Canada.

Kennedy G, Baran-Goldwax M, **Coutu-Nadeau E**, Pomerleau R, Knoth IS, Lippé S. Feeling everything: estrogen, cortisol, and sensory brain dynamics across the menstrual cycle. International Society of Psychoneuroendocrinology Annual Conference; 2025 Sep; New Orleans, LA, USA.

Herrero Babiloni A, Frimerman L, Fournier C, **Coutu-Nadeau E**, Ross A, Lavigne GJ, Martel MO. Sleep quality and prescription opioid misuse: the potential mediating role of pain intensity, negative affect, and opioid craving. *Sleep Med.* 2019 Dec;64(Suppl 1):S152–S153.

Frimerman L, Fournier C, **Coutu-Nadeau E**, Ross A, Martel MO. Determinants of prescription opioid craving and opioid-related hedonic feelings among patients with chronic pain prescribed opioid therapy. *J Pain.* 2019 Mar;20(4 Suppl): S55–S56.

Herrero Babiloni A, Frimerman L, Fournier C, **Coutu-Nadeau E**, Ross A, Lavigne GJ, Martel MO. Sleep quality and opioid craving among patients with chronic pain and prescribed opioid therapy: explorations of potential mediators. 11th Congress of the European Pain Federation; 2019 Sep; Valencia, Spain.

Herrero Babiloni A, Frimerman L, Fournier C, **Coutu-Nadeau E**, Ross A, Lavigne GJ, Martel MO. Sleep quality and prescription opioid misuse: the potential mediating role of pain intensity, negative affect, and opioid craving. World Sleep Congress; 2019 Sep; Vancouver, BC, Canada.

Frimerman L, Fournier C, **Coutu-Nadeau E**, Ross A, Martel MO. Determinants of prescription opioid craving and opioid-related hedonic feelings among patients with chronic pain prescribed opioid therapy. 23rd Annual McGill Pain Day; 2019 Jan; Montreal, QC, Canada.

Frimerman L, Fournier C, **Coutu-Nadeau E**, Ross A, Martel MO. Determinants of prescription opioid craving and opioid-related hedonic feelings among patients with chronic pain prescribed opioid therapy. 38th Annual Scientific Meeting of the American Pain Society; 2019 Apr; Milwaukee, WI, USA.

Ouellet N, Fournier C, Frimerman L, **Coutu-Nadeau E**, Martel MO. Factors associated with opioid craving among patients with chronic pain prescribed long-term opioid therapy. 14th Annual Research Day of McGill's Faculty of Dentistry; 2019 Apr; Montreal, QC, Canada.